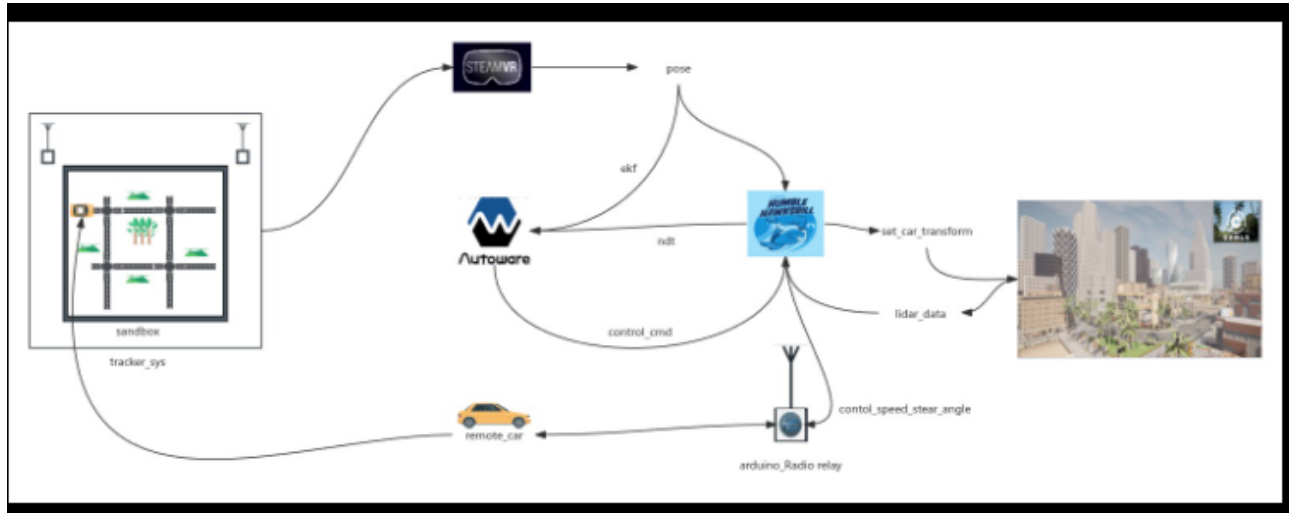


# this is used to record the state of logic of how to Dynamic obstacle avoidance, optimization and overall modeling

## 1. Summary of current situation



1.定位相关的补充信息如下：（1）steamvr追踪器一个置于车辆模型顶部，通过两个基站对其进行追踪，基站位于定位视野上部，无遮挡；（2）未测量steamvr的刷新速率，官方文档以查询的方式进行位姿获取，目前的查询频率为30帧/s；（3）定位误差为0.01m，但由于通过相似变换矩阵的方式进行了tracker坐标系与车辆的地图坐标系的转换，加上steamvr投影的曲面特性存在0.02-0.04m的位置误差。

（4）现实车模搭载轻量级imu但数据未使用，虚拟车模搭载lidar、camera、imu等多种传感器，定位时，以steamvr的追踪坐标作为输入融合lidar配准结果定位。2.控制系统的补充信息如下：（1）autoware的定位系统使用的是pid控制前后速度与mpc控制转向速度；实体车模使用pid控制，转向机构为pwm控制的三线舵机，该层级没有误差反馈机制（定位为运动状态反馈）（2）在ubuntu上监控autoware给出的控制指令，借助硬件nrf24L01发送给车模，通过自定义传输协议，32字节包，前11字节有效，包含速度和转动角速度两个关键参数，未进行延迟测量（请给出测量方案），粗估为50ms级别。（3）遥控车的转向、加速、制动等执行机构的响应时间未测量（请给出测量方案）。（4）控制数据的实时性未测量（请给出测量方案）3.状态估计与决策的补充信息如下：（1）steamvr的位姿数据通过autoware官方包的ekf算法融合了虚拟环境中获得的lidar信息进行定位，（2）车辆基于ackermann模型进行控制（autoware官方包）（3）基于autoware默认包进行规划，未进行其他优化（4）未测量整体数据处理时延（给出测量方案）

## 2. Delay measurement/System Testing

(1) tracker time delay use win + steamvr to test the refresh rate of the tracker

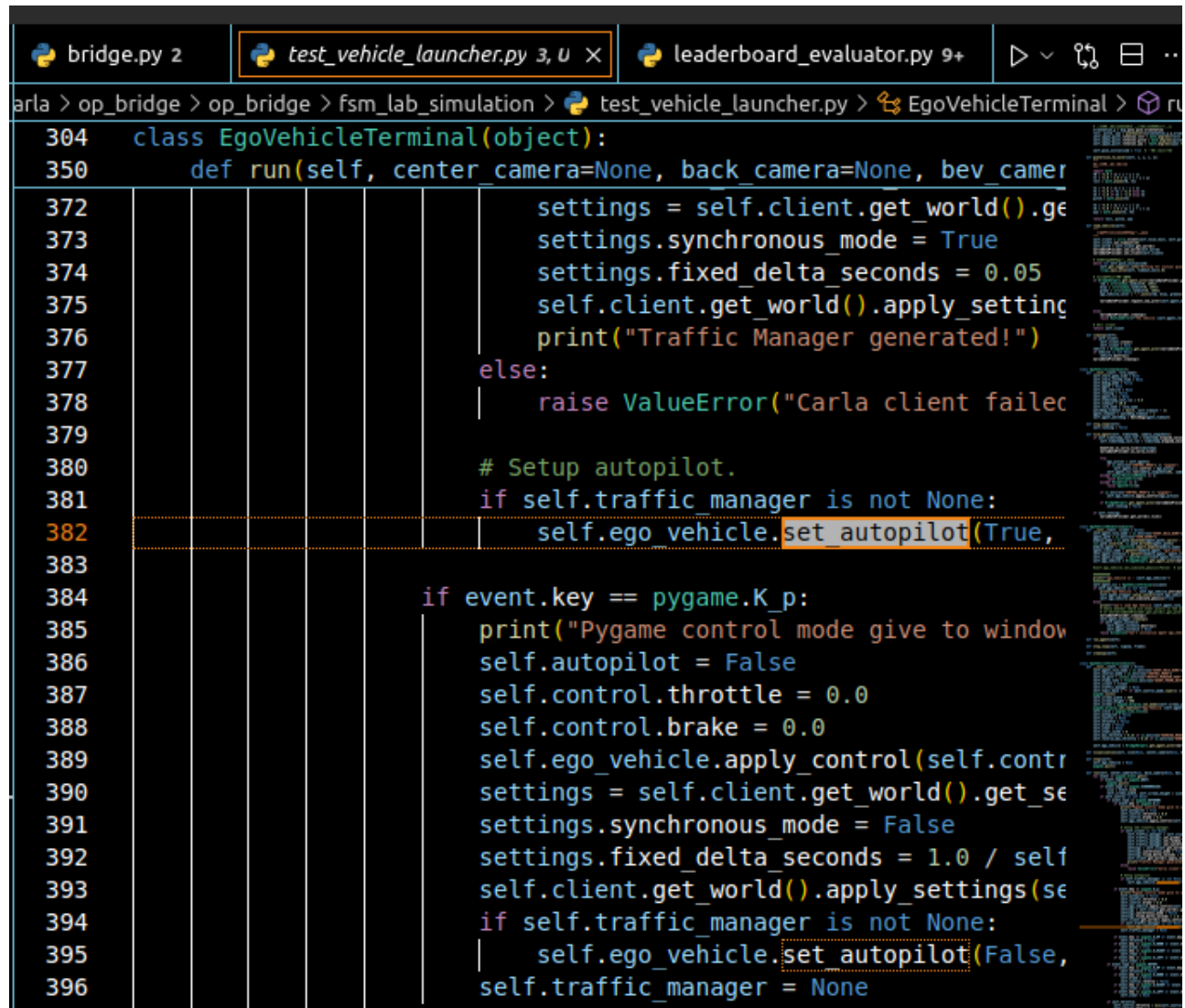
Vive Tracker 3.0 刷新率: 90.00 Hz

refresh rate of steamvr set by HMD is 90 need to design

## 3. Optimization tasks

(1). Positioning accuracy is stable to 0.01m using new Laser rangefinder to get accuracy coordinate (2). Delay is controllable or predictable/perceptible using win to test (question 2-1) (3). Overall mission

modeling (4). Model reduction RC Mosquito Car (5). sync step of carla set as sync mode and set fixed\_delta\_seconds as 1.0/30 (6). traffic controller vehicel has two kinds of controllers



```

304 class EgoVehicleTerminal(object):
350     def run(self, center_camera=None, back_camera=None, bev_camer
372         settings = self.client.get_world().ge
373         settings.synchronous_mode = True
374         settings.fixed_delta_seconds = 0.05
375         self.client.get_world().apply_setting
376         print("Traffic Manager generated!")
377     else:
378         raise ValueError("Carla client failed
379
380     # Setup autopilot.
381     if self.traffic_manager is not None:
382         self.ego_vehicle.set_autopilot(True,
383
384     if event.key == pygame.K_p:
385         print("Pygame control mode give to window
386         self.autopilot = False
387         self.control.throttle = 0.0
388         self.control.brake = 0.0
389         self.ego_vehicle.apply_control(self.contr
390         settings = self.client.get_world().get_se
391         settings.synchronous_mode = False
392         settings.fixed_delta_seconds = 1.0 / self
393         self.client.get_world().apply_settings(se
394         if self.traffic_manager is not None:
395             self.ego_vehicle.set_autopilot(False,
396             self.traffic_manager = None
  
```

4. Model Analysis Model and analyze the delay and control accuracy of the entire system to obtain a reasonable evaluation of the algorithm stability.